EXERCISE-I (Conceptual Questions) — Build Up Your Understanding

ALCOHOL

1. The compound A, B and C in the reaction sequence

$CH_3CH_2OH \xrightarrow{PBr_3} A \xrightarrow[\Delta]{alc.KOH} B \xrightarrow{Br_2} C$

are given by the set

(1) C_2H_5Br, CH_3CH_2OH, CH_3CHBr_2.

(2) C_2H_5Br, $CH\equiv CH$, $CH_2=CHBr$

(3) C_2H_5Br, $CH_2=CH_2$, $CH_2Br—CH_2Br$

(4) C_2H_5Br, CH_3CH_2OH, $BrCH_2—CH_2Br$

AE0001

2. Which of the following alcohols gives a red colour in Victor Meyer test

(1) $CH_3–CH_2–CH_2–OH$ (2) $CH_3–\underset{\displaystyle CH_3}{\underset{|}{CH}}–OH$

(3) $(CH_3)_3C–OH$ (4) $CH_3–\underset{\displaystyle OH}{\underset{|}{CH}}–CH_2–CH_3$

AE0003

3. Which of the following does not turn orange colour of chromic acid to green

(1) 1° alcohol (2) 2° alcohol

(3) 3° alcohol (4) Allyl alcohol

AE0005

4. p, s and t-alcohols can be distinguished by :-

(1) Reimer-Tiemann reaction

(2) Tollen's reagent

(3) Lucas test

(4) Lassaigne's test

AE0006

5. Consider the following reaction :

$C_2H_5OH + H_2SO_4 \rightarrow$ Product

Among the following, which one cannot be formed as a product under any conditions ?

(1) $C_2H_5OSO_3H$

(2) $H_2C=CH_2$

(3) $HC\equiv CH$

(4) $CH_3–CH_2–O–CH_2–CH_3$

AE0007

6. Select the incorrect option for the following reaction:

$CH_3–CH=CH_2 + H_2O \xrightarrow{H^+} CH_3–\underset{\displaystyle OH}{\underset{|}{CH}}–CH_3$

(1) This is an example of NAR of alkene

(2) In the first step, protonation of alkene takes place to form carbocation

(3) In the second step, *Nucleophilic attack of water* takes place on carbocation

(4) In the last step deprotonation takes place to form an alcohol

HC0201

7. For the reduction of aldehydes and ketones into alcohol the reagent which can be used is/are :

(1) H_2 in presence of Ni, Pt or Pd

(2) $NaBH_4$

(3) $LiAlH_4$

(4) All of these

CC0202

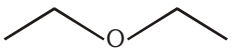
8. Which of the following does not reduces the carboxylic acids into alcohol ?

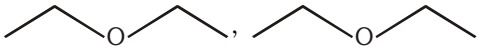
(1) $\xrightarrow[2.H_3O^+]{1.LiAlH_4/ether}$ (2) $\xrightarrow[2.H_3O^+]{1.B_2H_6}$

(3) $NaBH_4$ (4) $\xrightarrow[H^+]{ROH} \xrightarrow[Catalyst]{H_2}$

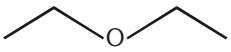
CA0203

9. $CH_3CH_2OH \xrightarrow[443K]{H_2SO_4} A$, $\xrightarrow[413K]{H_2SO_4} B$ A and B are (respectively)

(1)  , $CH_2=CH_2$

(2) 

(3) $CH_2=CH_2$, $CH_2=CH_2$

(4) $CH_2=CH_2$, 

AE0204

10. Select the correct option about denaturation of alcohol :
(1) It is the process of making commercial alcohol unfit for drinking
(2) $CuSO_4$ is added to give colour to alcohol
(3) Pyridine is added to give bed smell to alcohol
(4) All of these

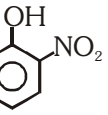
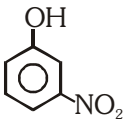
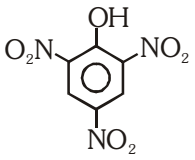
AE0205

11. Which of the following is insoluble in water ?
(1) Ethanol (2) Ethoxyethane
(3) Phenol (4) Pentane

AE0206

PHENOL

12. Nitration of phenol with conc. nitric acid gives :-

(1) 
(2) 
(3) 
(4) 

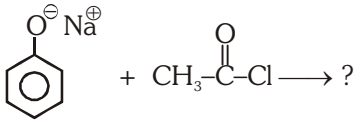
AE0008

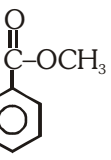
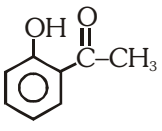
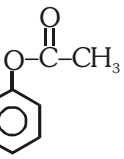
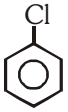
13. Deoxygenation of phenol can be achieved by distillation with :-
(1) Raney nickel
(2) Lithium aluminium hydride
(3) Sodium borohydride
(4) Zinc dust

AE0009

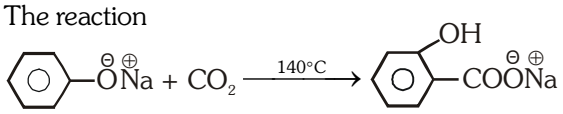
14. Which of the following compounds shows intramolecular hydrogen bonding :-
(1) p–Nitrophenol (2) Ethanol
(3) o–Nitrophenol (4) Methanamine

AE0010

15. 
Sodium phenate

(1) 
(2) 
(3) 
(4) 

CA0011

16. The reaction

is called :-
(1) Schotten Bauman reaction
(2) Kolbe Schmidt reaction
(3) Reimer–Tiemann reaction
(4) Lederer–Manasse reaction

AE0012

17. Phenol can be distinguished from ethanol by reactions with the following except :-
(1) Iodine and alkali
(2) Ferric chloride
(3) Acetyl chloride
(4) Bromine water

HD0013

18. Phenol on treatment with methyl chloride in the presence of anhydrous $AlCl_3$ gives chiefly :-
(1) o–cresol (2) m–cresol
(3) anisole (4) p–cresol

AE0014

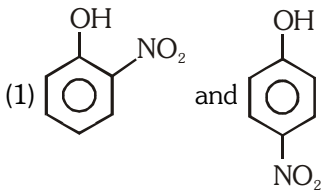
19. Phenol on heating with $NaNO_2$ and a few drops of conc. H_2SO_4 mainly gives :-
(1) p–nitrophenol
(2) p–nitrosophenol
(3) o–nitrophenol
(4) m–nitrosophenol

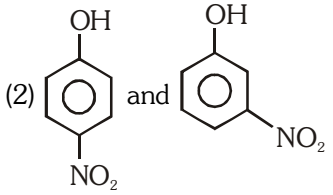
AE0015

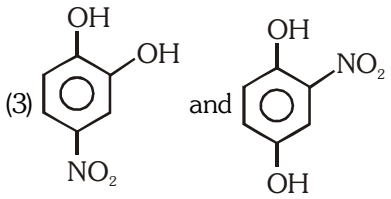
20. Phenol and benzoic acid are distinguished by :-
(1) Lucas reagent
(2) Victor Meyer test
(3) Caustic soda
(4) Sodium bicarbonate

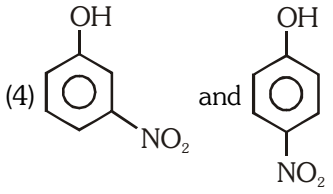
PO0016

21. Phenol on treatment with dil HNO_3 at low temp (298 K) gives two products P and Q. P is steam volatile but Q is not. P and Q are respectively.

(1)  and

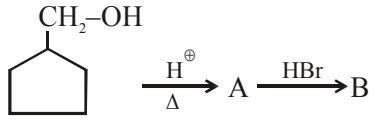
(2) 

(3) 

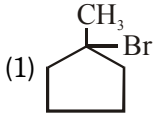
(4) 

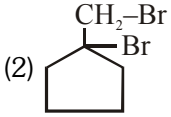
AE0207

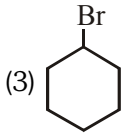
ETHER

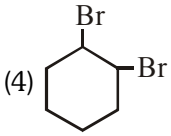
22. 

What is the structure of B :-

(1) 

(2) 

(3) 

(4) 

AE0018

23. $CH_3-\underset{\underset{OH}{|}}{CH}-CH_3 \xrightarrow{PBr_3} X \xrightarrow{Mg/ether} Y \xrightarrow{H_2O/H^+}$

The final product is :-

(1) $CH_3-\underset{\underset{CH_3}{|}}{CH}-OH$

(2) $CH_3-CH_2-CH_2-OH$

(3) $CH_3-\underset{\underset{CH_3}{|}}{CH}-O-\underset{\underset{CH_3}{|}}{CH}-CH_3$

(4) $CH_3-CH_2-CH_3$

AE0020

24. In the reaction sequence

$A \xrightarrow{HBr} B \xrightarrow{C_2H_5ONa}$ Ethoxyethane,

A and B are :-

(1) C_2H_6, C_2H_5Br

(2) CH_4, CH_3Br

(3) $CH_2=CH_2$, C_2H_5Br

(4) $CH\equiv CH$, $CH_2=CHBr$

AE0021

25. $CH_3-CH_2-OH + Ph-CH_2-OH \xrightarrow[140°C]{H^\oplus}$ of which is not obtained?

(1) $CH_3-CH_2-OCH_2-CH_3$

(2) $Ph-CH_2-OCH_2-Ph$

(3) $Ph-CH_2-O-CH_2-CH_3$

(4) $Ph-CH_2-O-CH_2-O-CH_3$

AE0023

26. Oxonium ion of ether has the structure :-

(1) $C_2H_5-O-\underset{\underset{CH_3}{|}}{CH}-\overset{\oplus}{O}-\underset{\underset{H}{|}}{H}$

(2) $CH_3-CH_2-\overset{\oplus}{\underset{\underset{H}{|}}{O}}-CH_2-CH_3$

(3) $(C_2H_5)_2O \rightarrow O$

(4) $CH_3-CH_2-O-CH_2-CH_2-\overset{\oplus}{\underset{\underset{H}{|}}{O}}-O-H$

AE0024

Z:\NODE02\B080-BA\TARGET\CHEM\ENG\MODULE-6\02 OXYGEN CONTAINING COMPOUND\02-EXE.P65

27. Which of the following does not react with aq. NaOH :-

(1) Phenol

(2) Benzoic acid

(3) CH_3COOH

(4) $CH_3–O–C_6H_5$

PO0026

28. A student tried two reactions for preparing tert-butyl ethyl ether :

(I) $C_2H_5ONa + CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-Cl \longrightarrow$

(II) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-ONa + CH_3-CH_2-Cl \longrightarrow$

Which reaction will give tert butyl ether ?

(1) Only I (2) Only II

(3) Both I & II (4) Neither I nor II

HD0208

CARBONYL COMPOUNDS

29. Acetaldehyde on warming with Fehling's solution gives a red precipitate of :-

(1) Elemental copper

(2) Cuprous oxide

(3) Cupric oxide

(4) Mixture of all of the above

CC0027

30. Acetone does not form :-

(1) A phenylhydrazone with phenylhydrazine

(2) A sodium bisulphite adduct with sodium bisulphite

(3) A silver mirror with Tollen's reagent

(4) An oxime with hydroxylamine

CC0028

31. CH_3CHO and CH_3COCH_3 can not be distinguished by :-

(1) Fehling solution (2) Grignard reagent

(3) Schiff's reagent (4) Tollen's reagent

CC0029

32. Acetone is obtained by the hydrolysis of the addition product of methyl magnesium iodide and :-

(1) HCHO (2) CH_3CHO

(3) CH_3COCH_3 (4) $CH_3–C\equiv N$

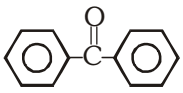
CC0031

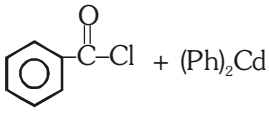
33. $Ph-\underset{\underset{Cl}{|}}{\overset{\overset{Cl}{|}}{C}}-CH_3 + \text{aq. KOH} \rightarrow A \xrightarrow[H^{\oplus}]{KCN} B$?

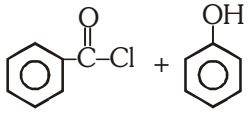
(1) 50% d + 50% ℓ (2) 80% d + 20% ℓ

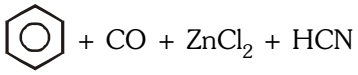
(3) Meso compound (4) optically active

CC0032

34.  can be obtained by :-

(1) 

(2) 

(3) 

(4) None of the above

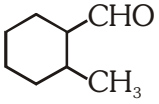
CC0033

35. Which does not react with $NaHSO_3$.

(1) $Ph-\underset{\underset{O}{\|}}{C}-H$

(2) $CH_3-O-C_6H_4-\underset{\underset{O}{\|}}{C}-H$

(3) $CH_3-C_6H_4-CH_2-\underset{\underset{O}{\|}}{C}-Ph$

(4) 

CC0034

36. Ketones can be prepared by :-

(1) Rosenmund reduction

(2) Etard reaction

(3) Cannizzaro reaction

(4) Friedel-Craft reaction

CC0035

37. Carbonyl compounds are best purified by :-

(1) Steam distillation

(2) Hydrolysis of sodium bisulphite adducts

(3) Fractional crytallisation

(4) Sublimation

CC0036

38. Carbonyl compounds readily undergo :-
(1) Nucleophilic substitutions
(2) Electrophilic addition reactions
(3) Nucleophilic addition reactions
(4) Free radical substitution reactions

CC0037

39. $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$ and $CH_3-\overset{\overset{O}{\|}}{C}-H$ are readily distinguished by their reaction with :-
(1) Iodine and alkali
(2) 2,4-dinitrophenylhydrazine
(3) Tollen's reagent
(4) All the above

CC0038

40. Which is the most suitable reagent for the following conversion

$$CH_3-CH=CH-CH_2-\overset{\overset{O}{\|}}{C}-CH_3 \rightarrow CH_3-CH=CH-CH_2-\overset{\overset{O}{\|}}{C}-OH$$

(1) $\xrightarrow[(ii)H^{\oplus}]{(i)\text{Tollen's reagent}}$ (2) $\xrightarrow[(ii)H^{\oplus}]{(i)\text{Benzoyal peroxide}}$

(3) $\xrightarrow[(ii)H^{\oplus}]{(i)I_2 \text{ and NaOH}}$ (4) $\xrightarrow{KMnO_4/H^{\oplus},\ \Delta}$

HD0040

41. Formaldehyde reacts with conc. alkali to form :-
(1) A resinous mass
(2) Formic acid
(3) A mixture of methanol and sodium formate
(4) Methanol

CC0041

42. Which of the following compounds does not give aldol condensation :-
(1) CH_3CHO (2) CH_3CH_2CHO
(3) HCHO (4) $CH_3CH_2CH_2CHO$

CC0042

43. Cannizzaro reaction is given by :-
(1) Aldehydes containing α-hydrogen atoms
(2) Aldehydes as well as ketones containing α-hydrogen atoms
(3) Aldehydes not containing α-hydrogen atoms
(4) Aldehydes containing β-hydrogen atoms

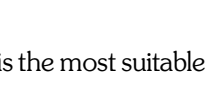
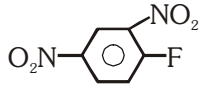
CC0043

44. Which of the following can be converted to $CH_3-CH=CH-CHO$:-
(1) Acetone (2) Acetaldehyde
(3) Propanaldehyde (4) Formaldehyde

CC0045

45. The product of reaction with primary amine and aldehyde is -
(1) $R-\overset{\overset{O}{\|}}{C}-OH$ (2) R–ONO
(3) R'–CH=N–R (4) $R-NO_2$

CC0046

46. Brady's reagent is
(1) $[Cu(NH_3)_4]SO_4$ (2) $KMnO_4/NaIO_4$
(3)  (4) 

CC0047

47. A compound with molecular formula C_3H_6O, not gives silver mirror with Tollen's reagent but forms oxime with hydroxyl amine. Compound will be -
(1) $CH_2=CH-CH_2-OH$
(2) CH_3CH_2CHO
(3) $CH_2=CH-O-CH_3$
(4) CH_3COCH_3

CC0048

48. Aldehyde and ketone are distinguished by reagent
(1) Fehling solution (2) H_2SO_4
(3) $NaHSO_3$ (4) NH_3

CC0049

49. Carbonyl group is converted into methylene group by -
(1) Acidic reduction
(2) Raney Ni
(3) Basic hydrolysis
(4) Normal Hydrogenation

CC0050

50. When acetaldol is treated with excess of acid then unsaturated product will be :-
(1) Alcohol (2) Aldehyde
(3) Acid (4) Alkyl halide

CC0051

51. The reagent used for the separation of acetaldehyde from acetophenone is -

(1) $NaHSO_3$ (2) $C_6H_5NHNH_2$

(3) NH_2OH (4) $NaOH + I_2$

CC0052

52. The most suitable reagent for the conversion of $RCH_2OH \longrightarrow RCHO$

(1) $KMnO_4$

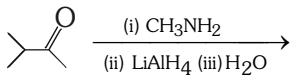
(2) $K_2Cr_2O_7$

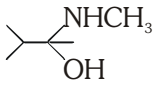
(3) CrO_3

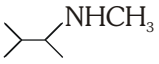
(4) PCC (Pyridinium chloro chromate)

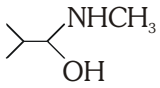
CC0053

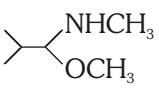
53. The major organic product formed from the following reaction is :–



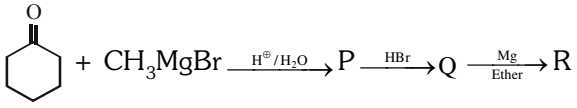
(1) 

(2) 

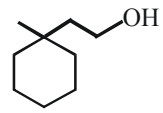
(3) 

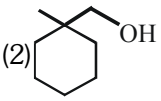
(4) 

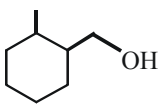
CC0054

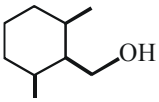
54. 



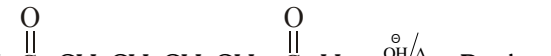
(1) 

(2) 

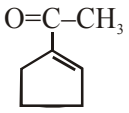
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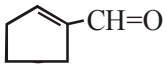
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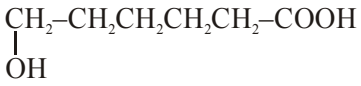
CC0056

55. 

(A) is :-

(1) 

(2) 

(3) 

(4) 

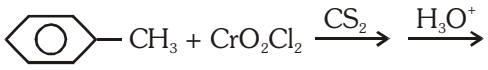
CC0057

56. Which of the following reaction will not give ketone?

(1) $R - MgX + \xrightarrow{R-COCl}$

(2) $R - CN + R - MgX \longrightarrow \xrightarrow{H_3O^+}$

(3) 

(4) 

CC0209

57. Select the incorrect option :

	Conversion	**Reagent**
(1)	Hexan-1-ol $\longrightarrow$ hexanal	$C_5H_5NH^+CrO_3Cl^-$
(2)	Ethanenitrile $\longrightarrow$ Ethanal	DIBAL–H
(3)	p-fluorotoluene, $\longrightarrow$ p-fluorobenzaldehyde	$K_2Cr_2O_7$, H^+
(4)	But-2-ene $\longrightarrow$ Ethanal	O_3/H_2O-Zn dust

CC0210

58. True statement about acetone is

(1) α– H of acetone is acidic due to strong electron withdrawing carbonyl group

(2) α–H of acetone is acidic due to resonance stabilisation of conjugate base

(3) It gives β-Hydroxy ketone with dilute alkali

(4) All

CC0211

Z:\NODE02\B080-BA\TARGET\CHEM\ENG\MODULE-6\02 OXYGEN CONTAINING COMPOUND\02-EXE.P65

59. $CH_3CH_3 \xrightarrow[h\nu]{Cl_2} (A) \xrightarrow{aq.KOH} (B) \xrightarrow{PCC} (C)$

$(D) \xleftarrow[\Delta]{OH^{\ominus}}$

The product D is :-

(1) $CH_2 = CH_2$

(2) $CH_3–\underset{\underset{OH}{|}}{CH}–CH_2–CHO$

(3) $CH_3–\underset{\underset{O}{||}}{C}–CH_3$

(4) $CH_3–CH=CH–CHO$

CC0212

60. Select the correct statement for C=O and C=C bond.

(1) Carbon–Oxygen double bond is polar but carbon-carbon double bond is non-polar

(2) Carbon–Oxygen bond length is 123 pm than that of carbon-carbon bond length is 134 pm

(3) carbonyl compounds undergo nucleophilic addition reaction but compounds containing ethylenic double bonds undergo electrophilic addition reaction

(4) All of these

CC0213

61. Select the structure of chromium complex formed by the reaction of toluene with chromylchloride followed by hydrolysis to give benzaldehyde and also the name of the reaction.

(1)  $CH(OCrCl_2)_2$ and Etard reaction

(2)  $CH(OCrOHCl_2)_2$ and Etard reaction

(3)  $CH_2(OCrOHCl_2)$ and Rosenmund reaction

(4)  $CH(OCrOH_2Cl)_2$ and Rosenmund reaction

CC0214

62. The methanal, ethanal and propanone are miscible with water because they form

(1) Vander waal's forces with water

(2) H-bond with water

(3) dipole-dipole bond with water

(4) ion-dipole bond with water

CC0215

63. The correct increasing order of carbonyl compounds towards nucleophilic addition reaction.

(1) Butanone < Propanone < Propanal < Ethanal

(2) Butanone < Propanal < Propanone < Ethanal

(3) Butanone < Ethanal < Propanone < Propanal

(4) Butanone < Ethanal < Propanal < Propanone

CC0216

64. Which of the following carbonyl group give the positive fehling test ?

(1) Aliphatic aldehydes (2) Aromatic aldehydes

(3) Ketones (4) Both (1) and (2)

PO0217

CARBOXYLIC ACID

65. When propanoic acid is treated with aqueous sodium bicarbonate, CO_2 is liberated. The C of CO_2 comes from :-

(1) methyl group

(2) carboxylic acid group

(3) methylene group

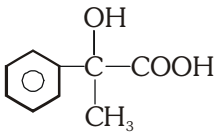
(4) bicarbonate

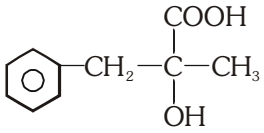
PO0060

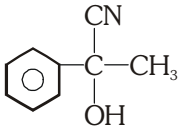
66. In a set of reactions acetic acid yielded a product D

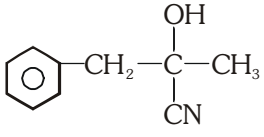
$$CH_3COOH \xrightarrow{SOCl_2} A \xrightarrow[Anhy.\ AlCl_3]{Benzene} B \xrightarrow{HCN} C \xrightarrow{H_3O^{\oplus}} D$$

The structure of D would be –

(1) 

(2) 

(3) 

(4) 

CA0061

ACID DERIVATIVE

67. The compounds A and B in the reaction sequence

$$B \xleftarrow{\text{Phenol}} CH_3COCl \xrightarrow{CH_3COONa} A$$

are given by the set :-

(1) $CH_3CO–O–COCH_3$, $C_6H_5CH_2OH$

(2) $CH_3CO–O–COCH_3$, $C_6H_5OCOCH_3$

(3) CH_3COCH_3, $C_6H_5OCOCH_3$

(4) $CH_3-\overset{\overset{O}{\|}}{C}-O-O-\overset{\overset{O}{\|}}{C}-CH_3$,

$CH_3-\overset{\overset{O}{\|}}{C}-O-C_6H_5$

CA0062

68. $CH_3–CH_2–COOH \xrightarrow{P/Cl_2} A \xrightarrow[KOH]{Alc.} B$

structure of B is :-

(1) $CH_2=CH–COOH$

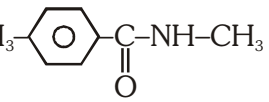
(2) $CH_3–\underset{\underset{Cl}{|}}{CH}–COOH$

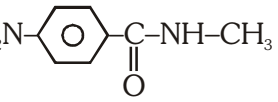
(3) $\underset{\underset{Cl}{|}}{CH_2}–CH_2–COOH$

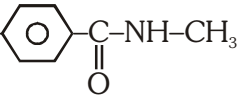
(4) $CH_3–CH_2–\overset{\overset{O}{\|}}{C}–Cl$

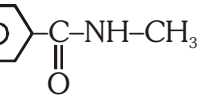
CA0063

69. Which is most reactive towards hydrolysis.

(1) 

(2) 

(3) 

(4) 

CA0065

70. Which of the following reagents may be used to distinguish between phenol and benzoic acid ?

(1) Victor-Mayer test

(2) Neutral $FeCl_3$

(3) Aqueous NaOH

(4) Tollen's reagent

PO0066

71. Acyl chlorides undergo :-

(1) Nucleophilic addition reactions

(2) Nucleophilic substitution reactions

(3) Electrophilic substitution reactions

(4) Electrophilic addition reactions

CA0067

72. The reaction of ethanol on acetic anhydride is an example of :-

(1) Nucleophilic addition

(2) Nucleophilic substitution

(3) Electrophilic addition

(4) Free radical substitution

CA0068

73. The reduction of acetamide gives :-

(1) $CH_3CH_2NH_2$

(2) $(CH_3)_2CHNH_2$

(3) $(CH_3)_3CNH_2$

(4) $(CH_3CH_2)_2NH$

CA0069

74. Which is used in preparation of aldehyde by rosenmund reduction

(1) Ester (2) Acid

(3) Acid halide (4) Alcohol

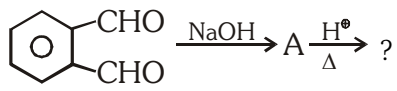
CC0070

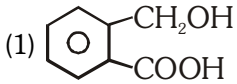
75. $CH_3-\overset{\overset{O}{\|}}{C}–NH_2 \xrightarrow[\Delta]{P_2O_5} ?$

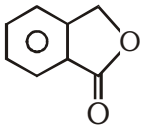
(1) CH_3COOH (2) $CH_3–CN$

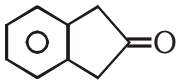
(3) $CH_3–CH_3$ (4) $CH_3–CHO$

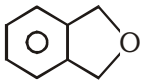
CA0072

76. 

(1) 

(2) 

(3) 

(4) 

CC0073

EXERCISE-I (Conceptual Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**3**	**1**	**3**	**3**	**3**	**1**	**4**	**3**	**4**	**4**	**4**	**4**	**4**	**3**	**3**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**2**	**3**	**4**	**2**	**4**	**1**	**3**	**4**	**3**	**4**	**2**	**4**	**2**	**2**	**3**
Que.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Ans.	**2**	**4**	**1**	**1**	**3**	**4**	**2**	**3**	**3**	**3**	**3**	**3**	**3**	**2**	**3**
Que.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	**3**	**4**	**1**	**1**	**2**	**1**	**4**	**2**	**2**	**2**	**4**	**3**	**4**	**4**	**4**
Que.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Ans.	**2**	**2**	**1**	**1**	**4**	**1**	**2**	**1**	**2**	**2**	**2**	**2**	**1**	**3**	**2**
Que.	76														
Ans.	**2**														

EXERCISE-III (Analytical Questions) — Master Your Understanding

1. Methanol can be distinguished from ethanol by the following except

(1) Reaction with iodine and alkali

(2) Reaction with salicylic acid and H_2SO_4

(3) Reaction with Lucas reagent

(4) Boiling point

AE0147

2. Ethanol on heating with acetic acid in the presence of a few drops of sulphuric acid gives the smell of

(1) Oil of wintergreen

(2) Oil of mustard

(3) An ester

(4) Oil of bitter almonds

AE0148

3. The compounds A, B and C in the reaction sequence are given by the set :-

$$CH_3CH_2OH \xrightarrow[\Delta]{HBr} A \xrightarrow{Mg} B \xrightarrow[(ii)\,H_3O^{\oplus}]{(i)\,CH_3CHO} C$$

(1) CH_3CH_2Br, CH_3CH_2MgBr, $(CH_3)_3C—OH$

(2) CH_3CH_2Br, $(CH_3CH_2)_2Mg$, $(CH_3)_2CHCH_2OH$

(3) CH_3CH_2Br, CH_3CH_2MgBr, $CH_3CH(OH)CH_2CH_3$

(4) CH_3CHBr_2, $CH_3CH(MgBr)_2$, $CH_3CH(OH)CH_3$

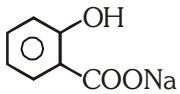
AE0149

4. Which of the following reactions will not lead to a phenol :-

(1) $C_6H_5SO_3Na + NaOH \xrightarrow[ii)\,H^+]{i)\,Fuse/\Delta}$

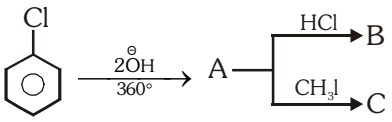
(2) $C_6H_5N_2Cl + H_2O \xrightarrow{Boil}$

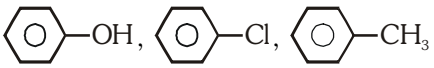
(3) $C_6H_5ONa + RX \xrightarrow[Heat]{NaOH}$

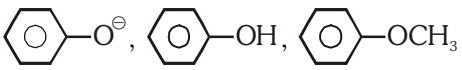
(4)  + NaOH(CaO) $\xrightarrow{Heat}$

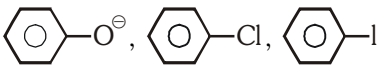
HD0150

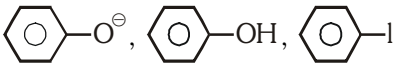
5. The structures of the compounds / ions A, B and C in the reaction sequence are given by the set :-



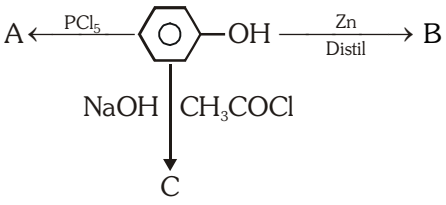
(1) 

(2) 

(3) 

(4) 

HD0151

6. 

The compounds A, B and C in the above reaction sequence are :-

(1) Chlorobenzene, benzene, methyl benzoate

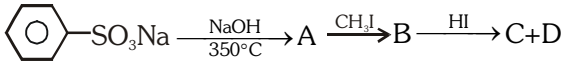
(2) Triphenyl phosphate, benzene, phenyl acetate

(3) Benzyl chloride, benzene, phenyl acetate

(4) Benzyl chloride, benzene, phenylacetyl chloride

AE0152

7. In the reaction sequence —



A, B, C and D are given by the set :-

(1) Sodium phenate, anisole, C_6H_5I, CH_3OH

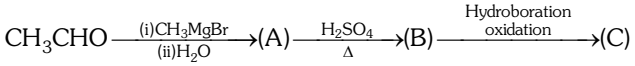
(2) Sodium phenate, phenetole, C_2H_5I, C_6H_5OH

(3) Sodium phenate, anisole, C_6H_5OH, CH_3I

(4) Sodium phenate, phenetole, C_6H_5I, C_2H_5OH

HD0153

8. Compound A and C in the following reaction are



(1) Identical

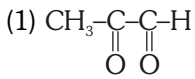
(2) Functional isomer

(3) Positional isomer

(4) Optical isomer

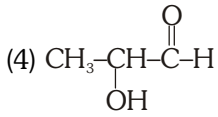
CC0154

9. Which of the following aldehydes does not form iodoform on heating with iodine and alkali :-

(1) 

(2) ICH_2CHO

(3) $CH_3–CH_2–CHO$

(4) 

HD0155

10. A carbonyl compound gives a positive iodoform test but does not reduce Tollen's reagent or Fehling's solution. It forms a cyanohydrin with HCN, which on hydrolysis gives a hydroxy acid with a methyl side chain. The compound is :-

(1) Acetaldehyde

(2) Propionaldehyde

(3) Acetone

(4) Crotonaldehyde

CC0157

11. Which of the following statement is wrong :-

(1) All methyl ketones give a positive iodoform test

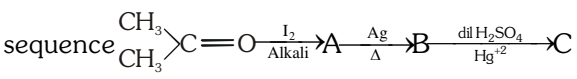
(2) Acetaldehyde is the only aldehyde that gives iodoform test

(3) All secondary alcohols give positive iodoform test

(4) Any alcohol that can be oxidised to an acetyl group gives a positive iodoform test

HD0158

12. The compounds A, B and C in the reaction

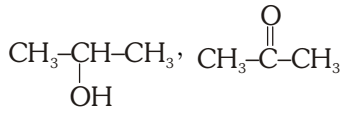
sequence 

are given by the set :-

(1) CHI_3, $H_2C=CH_2$, $CH_3CH_2–OH$

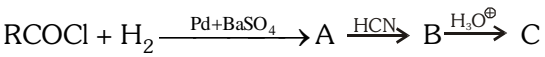
(2) CHI_3, $HC\equiv CH$, CH_3CHO

(3) CHI_3, $CH_3–C\equiv CH$, CH_3COCH_3

(4) CHI_3, 

HD0159

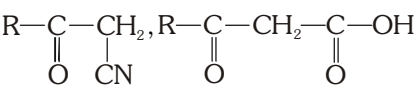
13. In the reaction sequence



A,B and C are given by the set :-

(1) RCHO, $RCH(OH)CN$, $RCH(OH)CH_2NH_2$

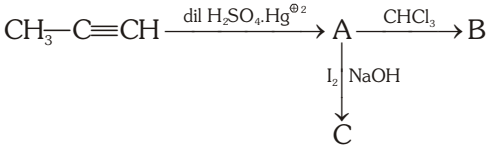
(2) RCHO, RCH(OH)CN, RCH(OH)COOH

(3) RCHO, 

(4) RCHO, $R–CH_2–CN$, $R–CH_2–COOH$

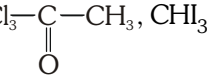
CA0160

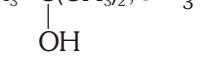
14. In the reaction sequence

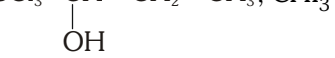


A, B and C are given by the set :-

(1) CH_3CH_2CHO, $CH_3CH_2CH_2Cl$, CHI_3

(2) CH_3COCH_3,  , CHI_3

(3) CH_3COCH_3,  , CHI_3

(4) CH_3CH_2CHO,  , CHI_3

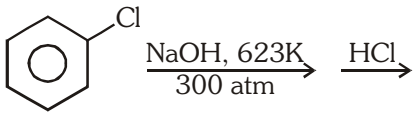
CC0161

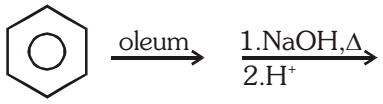
15. The reagents A and B in the reaction sequence

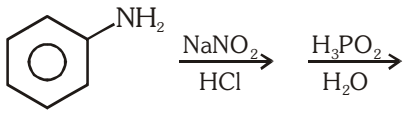
$CH_3COOC_2H_5 \xrightarrow{A} CH_3COOC(CH_3)_3$

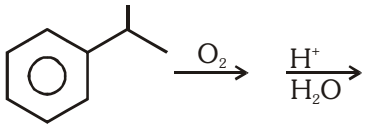
$\quad\quad\quad

24. Which of the following reaction sequence does not give phenol ?

(1) 

(2) 

(3) 

(4) 

AE0222

25. Which of the following reactions does not give aldehyde as major product ?

(1) $RCH_2OH \xrightarrow[N]{CrO_3\cdot} HC\ell$

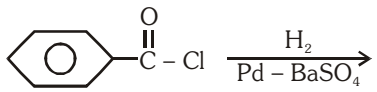
(2) $RCH_2OH \xrightarrow[CrO_3]{anhy.}$

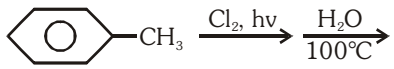
(3) $RCH_2OH \xrightarrow[573K]{Cu}$

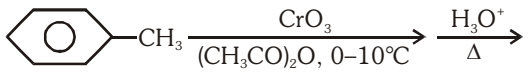
(4) $RCH_2OH \xrightarrow{KMnO_4,\ H^+}$

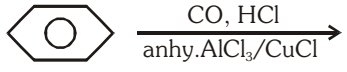
CC0223

26. Which of the following can be used to prepare benzaldehyde ?

(I) 

(II) 

(III) 

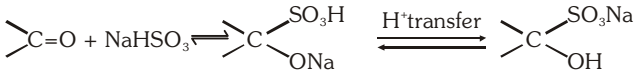
(IV) 

(1) I, II (2) II, III, IV

(3) I, III, IV (4) I, II, III, IV

CC0224

27. Select the correct option for the following reaction:



(1) The position of equilibrium is on RHS for most ketones

(2) The position of equilibrium is on LHS for most aldehydes

(3) The hydrogen sulphite addition compound is water insoluble

(4) Sulphite addition compounds are useful for separation and purification of aldehydes

CC0225

28. Select the correct option about Fehling's test :

(1) Fehling test is given by all aldehydes but not by ketones

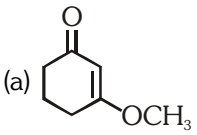
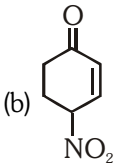
(2) A red brown ppt. of CuO confirms the test

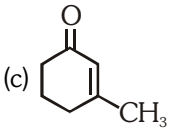
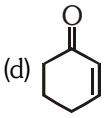
(3) Fehling solution 'A' is aq. $CuSO_4$

(4) Fehling solution 'B' is alkaline sodium potassium citrate

PO0226

29. Correct reactivity order towards nucleophilic addition is

(a)  (b) 

(c)  (d) 

(1) b > a > d > c (2) a > b > d > c

(3) b > d > c > a (4) b > d > a > c

CC0227

30. $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-OH$ is obtained from following method

(1) $CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-Cl + H_2O$

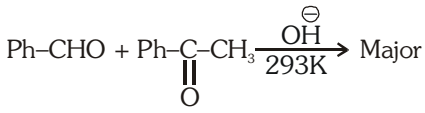
(2) $CH_3CHO, CH_3-CH_2MgBr, H_2O$

(3) $H-CHO, CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}\,Mg\,Br, H_2O$

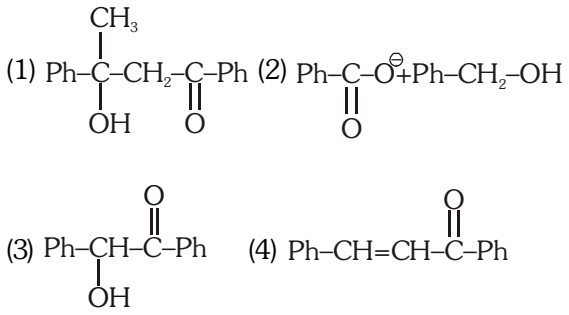
(4) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-NH_2+HNO_2$

CC0228

31. For the reaction :-



The major product is :



CC0229

32. Which of the following substrates given the same product on the reduction with DIBAL-H ?

(1) CH_3–$(CH_2)_4$–CN and CH_3–$(CH_2)_5$–COOH

(2) CH_3–$(CH_2)_4$–CN and CH_3–$(CH_2)_4$–$COOC_2H_5$

(3) CH_3–$(CH_2)_4$–COOH and CH_3–$(CH_2)_4$–$COCH_3$

(4) CH_3–$(CH_2)_5$–COOH and $CH_3(CH_2)_4$–$COOC_2H_5$

AM0230

33. Choose the correct alternative from the following.

(1) Ketones are more reactive than aldehydes towards NAR.

(2) Aldehydes are more reactive than ketones towards NAR.

(3) Formaldehyde is the least reactive carbonyl Compound towards NAR

(4) Steric hindrance does not play a role to effect the reactivity of carbonyl componds towards NAR.

CC0231

34. Which of the following reagents is/are used in the given reaction ?

RCHO $\longrightarrow$ RCOOH

(1) $KMnO_4/H^+$ (2) Potassium dichromate

(3) Tollen's reagent (4) All of the above

CC0232

35. Whinch of the following statements is/are correct?

(1) Aldehydes are generally oxidised under vigorous conditions

(2) Ketones are easily oxidised to carboxylic acids even under mild oxidising agents

(3) Oxidation of ketones involves carbon-carbon bond cleavage to give a mixture of carboxylic acids having lesser number of c-atoms than the parent ketones.

(4) All of the above

CC0233

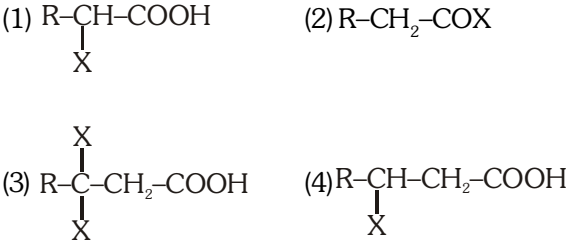
36. Which type of reaction(s) is/are involved in the cannizzaro reaction ?

(1) Reduction (2) Oxidation

(3) Both (1) and (2) (4) None of these

CC0234

37. The product formed during Hell-volhard-Zelinsky reaction is :-



CA0235

EXERCISE-III (Analytical Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**3**	**3**	**3**	**3**	**2**	**2**	**3**	**3**	**3**	**3**	**3**	**2**	**2**	**3**	**3**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**3**	**1**	**1**	**4**	**2**	**4**	**2**	**3**	**3**	**4**	**4**	**4**	**3**	**3**	**3**
Que.	31	32	33	34	35	36	37								
Ans.	**4**	**2**	**2**	**4**	**3**	**3**	**1**								

Z:\NODE02\B080-BA\TARGET\CHEM\ENG\MODULE-6\02 OXYGEN CONTAINING COMPOUND\02-EXE.P65